Somaiya AgentReel Challenge - Process Report

Team T13 | Brief B06 "Why Somaiya?" | LY Agentic AI Laboratory | Autonomy level: L4

1. Brief interpretation and creative strategy

The brief asks for a 9:16, ~45 s reel for engineering aspirants that answers "Why Somaiya?". Audience insight: aspirants and parents distrust inflated claims, so the strategy is trust through verifiable facts rather than hype. Hook: the topic is stated in the first 3 seconds (on-screen title plus the opening voice-over line). Tone: energetic and inspiring. Call to action: explore the official website, shown on the end card together with the logo and the line "Created with AI agents - LY Agentic AI Lab". Visuals use only official media-kit photos, so no real or generated faces are created.

2. Agent graph, framework and models

A plain-Python orchestrator (run.py) chains seven agents with JSON hand-offs: Planner, Researcher (RAG), Scriptwriter, Storyboarder, Asset Generator, Editor and Critic. A custom orchestrator was chosen over a framework to keep every hand-off, log line and retry visible and easy to reproduce. Each agent exists for one reason: the Planner turns the brief into stages and success criteria; the Researcher is the only agent allowed to introduce facts; the Scriptwriter and Storyboarder turn those facts into timed narration and shots; the Asset Generator and Editor produce audio, captions, shots and the final MP4; the Critic checks the result and routes failures back. Model routing: Gemini (hosted, free tier) is called first for all language stages; a local Ollama model (qwen2.5:3b) is the automatic fallback when the API is rate-limited. Voice-over uses Edge-TTS, and editing uses MoviePy and ffmpeg (loudnorm to about -14 LUFS). The diagram is in docs/agent_graph.png.

3. Grounding and hallucination control

The official brochure is split into chunks with page numbers and retrieved per brief. The Researcher must return, for every fact, a source id and a verbatim quote. Code then checks that the quote really occurs in that chunk; facts that fail are moved to a rejected list and never reach the script. The Critic additionally blocks any number in the script that is absent from the verified facts, and any unverified ranking, placement or package wording.

4. Time and cost per stage

Stage	Tool / model	Cost (INR)	Notes
Planner	Gemini (free tier)	0	Writes plan.json
Researcher	Gemini + local retrieval	0	Quote-verified facts.json
Scriptwriter	Gemini (free tier)	0	Word budget about 2.3 words/s
Storyboarder	Gemini (free tier)	0	6-8 shots, durations rescaled
Asset Generator	Edge-TTS, Pillow	0	Voice, captions (.srt), shots, end card
Editor	MoviePy + ffmpeg	0	1080x1920, 30 fps, H.264/AAC
Critic	Rule-based checks	0	12 checks, qa_report.json

Total API spend per run is INR 0 (free tier and local tools), well under the INR 500 budget. Per-call latency, tokens and provider are recorded in logs/run_log.jsonl and logs/cost_report.csv. Two complete end-to-end runs were completed and both passed QA.

5. Critic loop: what failed and what was fixed

- Hallucination filter: when the weak local fallback model answered during an API rate limit, it returned facts without source quotes. The Researcher rejected all 7 of them automatically, so none entered the script.
- Critic gap found in testing: an early run passed QA with 0 retrieved facts, because checking an empty list counts as a pass. We added explicit checks (at least 3 facts, at least 1 body segment, every segment cites a fact) so such a run now fails and loops back.
- Source-id formatting: the model returned ids in square brackets, which broke matching and discarded valid facts. Ids are now normalised before verification.
- The Critic can loop back up to 3 times with written feedback (for example, shorten the script if the video exceeds 60 s). In the final runs all checks passed on the first attempt.

6. Human interventions

No human edits were made to any generated file during the runs; the human only started the command, so the level is L4. Interventions during development were code fixes, not output edits. To remove remaining manual work we would add automatic API-quota detection, an automatic check that a fallback model produced valid JSON, and a vision check on frames.

7. Limitations, ethics and improvements

- The quote check proves a fact exists in the brochure, not that the model interpreted it correctly; a human fact-check of the script is still advised.
- Gemini does not honour the seed, so two runs with seed 42 differ slightly; the local model is reproducible but weaker.
- Edge-TTS and Gemini need internet access; a network drop stops a run (retry logic added for TTS). Free-tier quotas can force the fallback model.
- Ethics: only official photos and facts are used, no real faces or voices are generated, a synthetic voice is declared, and the AI-disclosure line appears on the end card.
- Next steps: caption breaks at natural phrase boundaries, richer facts via a larger retrieval window, and multi-critic voting on claims.